

Program 6: Deploy an Automation build pipeline using Dockerhub 21/11/2025

Project Structure:

```
prog6
|- Dockerfile
|- server.js
|- package.json
```

Steps for Execution:

Step 1: Move to the directory where we work for DevOps lab. Then create a project directory with the name prog1

```
> mkdir prog6
```

```
> cd prog6/
```

```
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~$ cd devops_lab/
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab$ mkdir prog6
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab$ cd prog6/
```

Step 2: Inside the prog6 folder have 2 files in it,

1. Dockerfile
2. server.js

The content which should be included in that file is shown in the below image

```
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog6$ nano Dockerfile
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog6$ cat Dockerfile
# Use an official lightweight Node image
FROM node:18-alpine

# Create app directory
WORKDIR /usr/src/app

# Copy package.json and package-lock if present
COPY package*.json ./

# Install app dependencies
RUN npm install --production

# Copy app source
COPY . .

# Expose port (same as server listens on)
EXPOSE 3000

# Start command

1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog6$ nano server.js
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog6$ cat server.js
// server.js - simple web server
const http = require('http');

const PORT = process.env.PORT || 3000;

const requestHandler = (req, res) => {
  res.writeHead(200, { 'Content-Type': 'text/plain' });
  res.end('Hello from Program6 container!\\n');
};

const server = http.createServer(requestHandler);

server.listen(PORT, () => {
  console.log(`Server running on port ${PORT}`);
});
```

Step 3: Do the npm initialization by the command,

> npm init -y

This above command adds the package.json into the project folder

Then edit the current package.json in such a way that it should contain the content which is shown in the below image

```
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog6$ npm init -y
Wrote to /home/chinnu/devops_lab/prog6/package.json:

{
  "name": "prog6",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC"
}

1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog6$ nano package.json
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog6$ cat package.json
{
  "name": "prog6",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "scripts": {
    "start": "node server.js",
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC"
}
```

Step 4: The final project show contain the files which are mentioned in the below image

```
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog6$ ls
Dockerfile  package.json  server.js
```

Step 5: Now build and tag the image by using the commands,

> docker build -t prog6 .

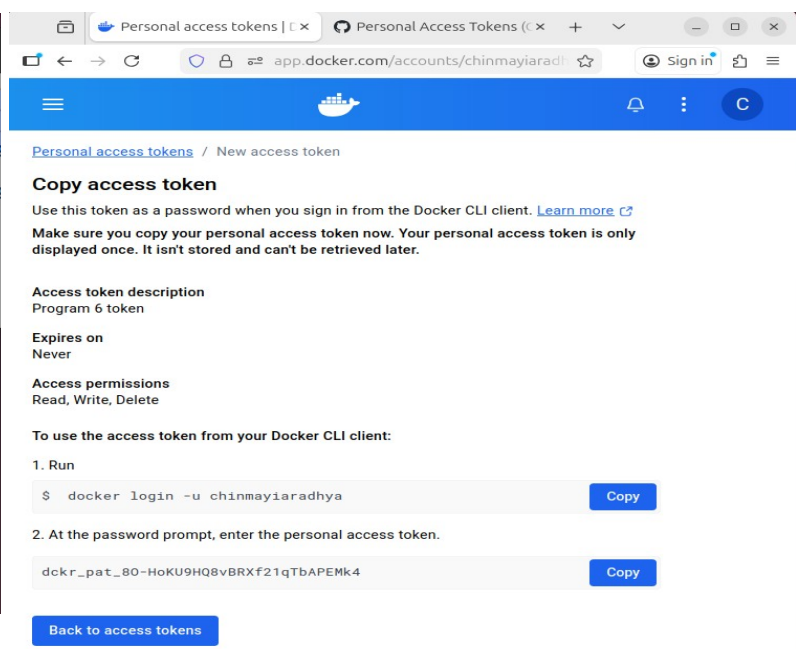
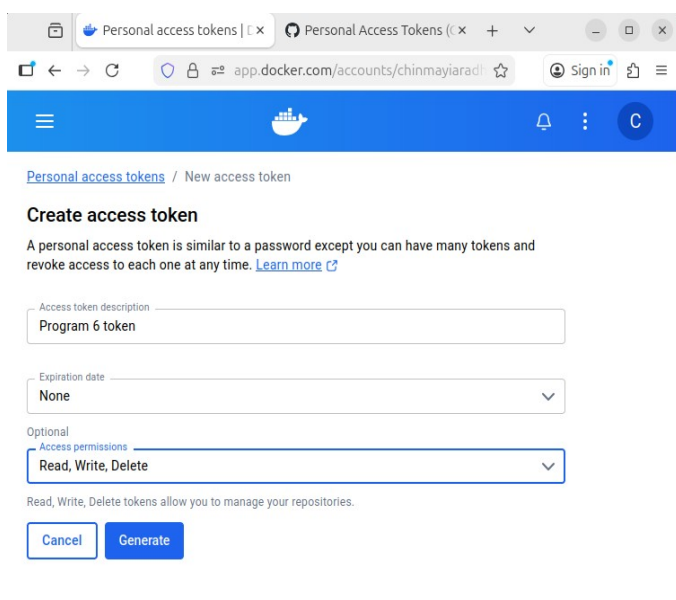
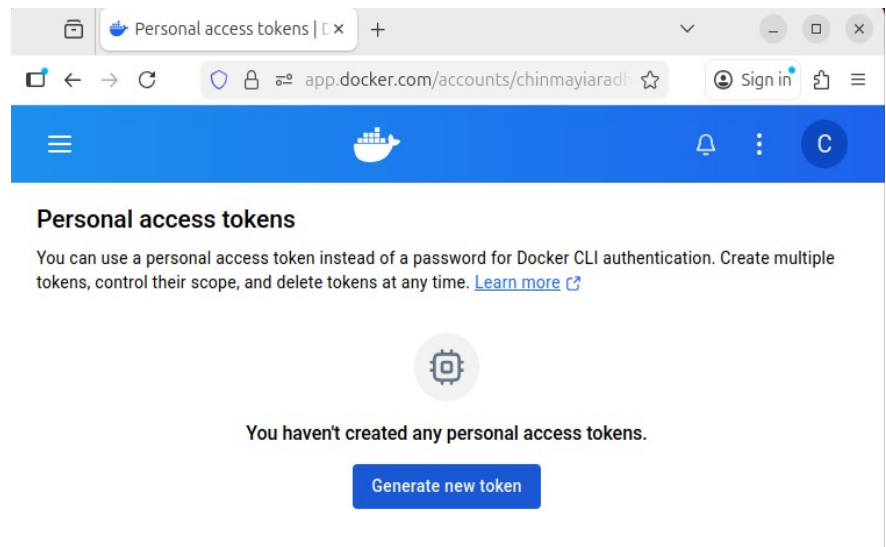
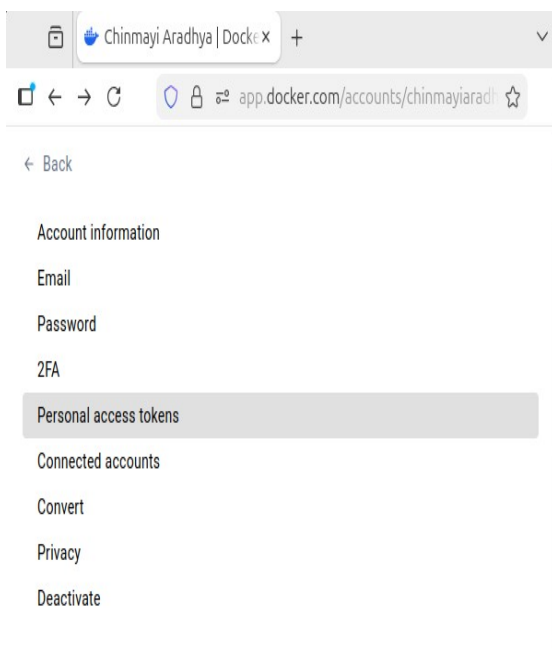
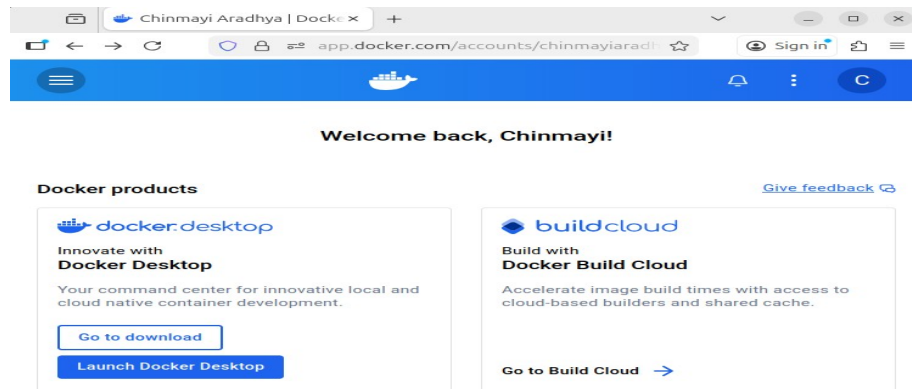
> docker tag prog6 <dh-username>/prog6:latest

```
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog6$ docker build -t prog6 .
[+] Building 6.1s (11/11) FINISHED                    docker:default
=> [internal] load build definition from Dockerfile    0.0s
=> => transferring dockerfile: 399B                  0.0s
=> [internal] load metadata for docker.io/library/node:18-alpine 3.4s
=> [auth] library/node:pull token for registry-1.docker.io 0.0s
=> [internal] load .dockerignore                      0.0s
=> => transferring context: 2B                        0.0s
=> CACHED [1/5] FROM docker.io/library/node:18-alpine@sha256:8d6421d663b 0.0s
=> => resolve docker.io/library/node:18-alpine@sha256:8d6421d663b4c28fd3 0.0s
=> [internal] load build context                      0.0s
=> => transferring context: 1.11kB                    0.0s
=> [2/5] WORKDIR /usr/src/app                        0.0s
=> [3/5] COPY package*.json ./                      0.0s
=> [4/5] RUN npm install --production                1.9s
=> [5/5] COPY . .                                    0.1s
=> exporting to image                                0.5s
=> => exporting layers                                0.2s
=> => exporting manifest sha256:338694a11f3f20291ba3f5661b4f73c017bc0602 0.0s
=> => exporting config sha256:a24ab2c5de9c4f9d5df77be30627615a9e05ab0c16 0.0s
=> => exporting attestation manifest sha256:9f6ca74b3d41bbf27e43b4608742 0.0s
=> => exporting manifest list sha256:53b169be1f8234c35609c62c40dfe8676fc 0.0s
=> => naming to docker.io/library/prog6:latest        0.0s
=> => unpacking to docker.io/library/prog6:latest     0.1s
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog6$ docker tag prog6 chinmayiaradhy/prog6:latest
```

DOCKER HUB

Step 6: Open the browser and search for Docker hub, login with the specific credentials email and password. Then create a Personal Access Token (PAT). Follow the below steps:

> login to docker hub > settings > Personal Access Tokens > Generate new token > fill the description and add the permission of read, write and delete > Generate > Copy the command which will be given to login and also copy the password



Step 7: Come back to the before terminal and login to the docker hub with the login command given in the browser and push the tagged image by using the commands,

> docker login -u <dh-username>

//it will ask for the password, paste the PAT which has been created in the step 5 here

> docker push <dh-username>/prog6:latest

```
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog6$ docker login -u chinmayiaradhya
```

Info → A Personal Access Token (PAT) can be used instead.
To create a PAT, visit <https://app.docker.com/settings>

Password:

Login Succeeded

```
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog6$ docker push chinmayiaradhya/prog6:latest
```

The push refers to repository [docker.io/chinmayiaradhya/prog6]

25ff2da83641: Pushed

f18232174bc9: Pushed

8704486a9f75: Pushed

1e5a4c89cee5: Pushed

ccfd53acbeba: Pushed

dd71dde834b5: Pushed

198b87eb3618: Pushed

68a12e8610d6: Pushed

93a6d445d29b: Pushed

Step 8: Verify whether the repository has created by the image in the browser of docker hub,

The screenshot shows the Docker Hub interface for the repository 'chinmayiaradhya/prog6'. The left sidebar contains navigation links: Repositories, Hardened Images, Collaborations, Settings, Default privacy, Notifications, Billing, Usage, Pulls, and Storage. The main content area shows the repository details, including the name 'chinmayiaradhya/prog6', the last push time '3 minutes ago', and a table of tags. The 'latest' tag is highlighted, showing it was pushed 3 minutes ago by 'chinmayiaradhya'. Below the tag information, there is a table with columns: TAG, Digest, OS/ARCH, and Last pull. The 'latest' tag has a digest of '338694a11f3f', is for 'linux/amd64', and was pulled 'less than 1 day' ago.

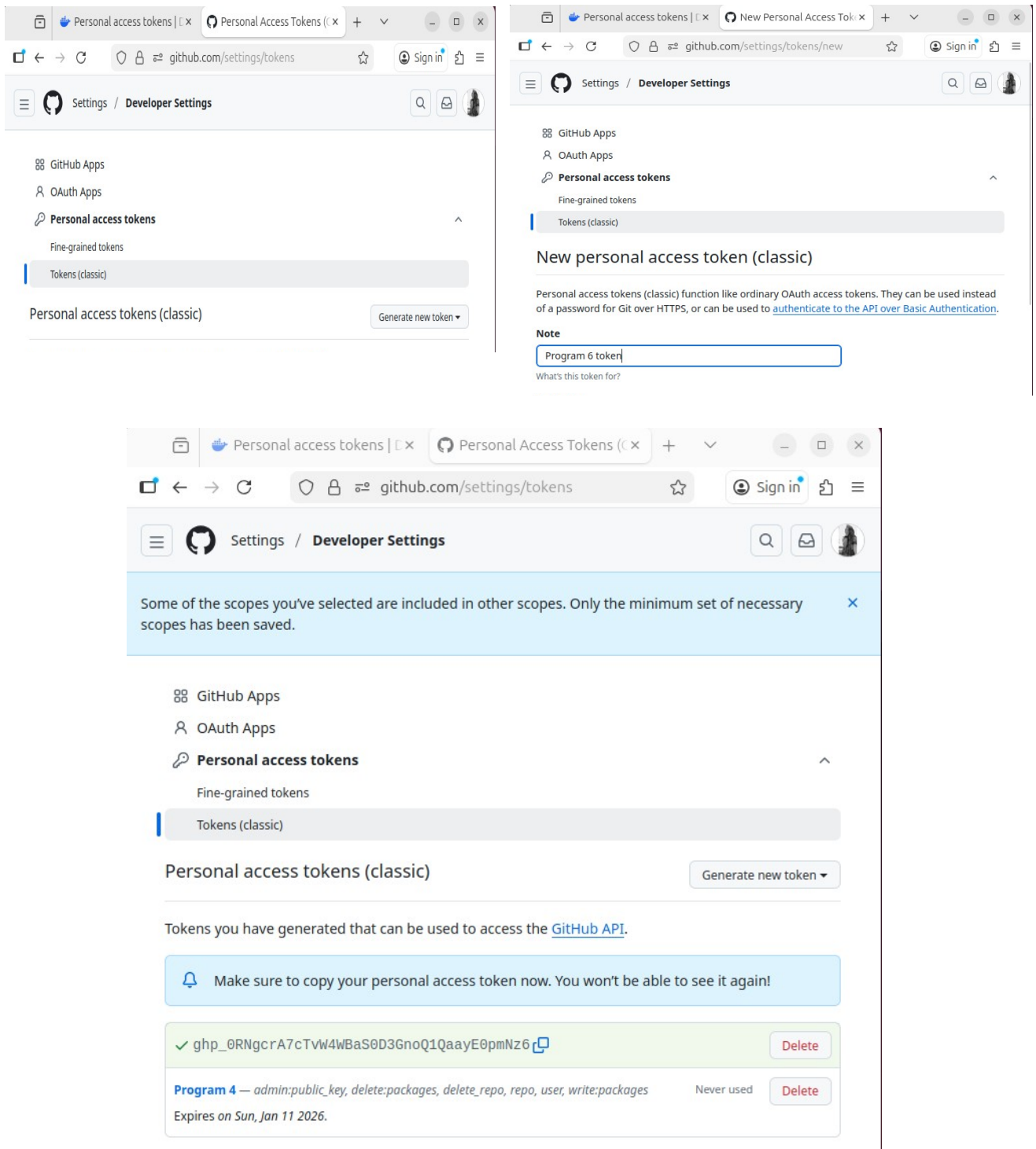
TAG	Digest	OS/ARCH	Last pull
latest	338694a11f3f	linux/amd64	less than 1 day

GITHUB

Step 9: Open the browser and search for github, login with the specific credentials email and password. Then create a Personal Access Token (PAT). Follow the below steps:

> login to github > settings > developer settings > Personal Access Token > Tokens (Classic) >

Generate new token > specify the note and select the scope for the PAT > Generate > Copy the PAT which will be displayed after clicking the generate



Step 10: Now login to ghcr by using the command,

```
> echo <PAT-of-github> | docker login ghcr.io -u <gh-username> --password-stdin
```

```
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/p
rog6$ echo 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 | docker login ghcr.io -u Chinmayi
-Aradhya --password-stdin
```

WARNING! Your credentials are stored unencrypted in '/home/chinnu/.docker/config.json'.
Configure a credential helper to remove this warning. See
<https://docs.docker.com/go/credential-store/>

Login Succeeded

Step 11: Tag the image which has created in step 4 and push to ghcr by using the commands,

```
> docker tag prog6 ghcr.io/<gh-username>/prog6:latest
```

```
> docker push ghcr.io/<gh-username>prog6:latest
```

```
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog6$ docker tag prog6 ghcr.io/chinmayi-aradhya/prog6:latest
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog6$ docker push ghcr.io/chinmayi-aradhya/prog6:latest
The push refers to repository [ghcr.io/chinmayi-aradhya/prog6]
f18232174bc9: Pushed
8704486a9f75: Pushed
25ff2da83641: Pushed
93a6d445d29b: Pushed
dd71dde834b5: Pushed
68a12e8610d6: Pushed
1e5a4c89cee5: Pushed
198b87eb3618: Pushed
ccfd53acbeba: Pushed
latest: digest: sha256:53b169be1f8234c35609c62c40dfe8676fcc8d4f1c8090d11dd2ad2d008f576d size: 856
```

Step 12: Create a repository in github with the name prog6

Create a new repository

Repositories contain a project's files and version history. Have a project elsewhere? [Import a repository](#).
Required fields are marked with an asterisk (*).

1

General

Owner *

Chinmayi-Aradhya

Repository name *

prog6

pro is available.

Great repository names are short and memorable. How about [redesigned-octo-lamp](#)?

Description

Program 6

9 / 350 characters

2

Configuration

Choose visibility *

Public

Choose who can see and commit to this repository

Add README

READMEs can be used as longer descriptions. [About READMEs](#)

off

Add .gitignore

.gitignore tells git which files not to track. [About ignoring files](#)

No .gitignore

Add license

Licenses explain how others can use your code. [About licenses](#)

No license

Create repository

Step 13: Now initialize the git and perform the initial commit by using the below commands,

```
> git init
> git add .
> git commit -m "Initial Commit for Program6"
> git branch -M main
> git remote add origin https://github.com/<gh-username>/prog6.git
```

```

1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog6$ git init
git add .
git commit -m "Initial Commit for Program6"
git branch -M main
git remote add origin https://github.com/Chinmayi-Aradhya/prog6.git
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /home/chinnu/devops_lab/prog6/.git/
[master (root-commit) f2541c6] Initial Commit for Program6
 3 files changed, 48 insertions(+)
 create mode 100644 Dockerfile
 create mode 100644 package.json
 create mode 100644 server.js

```

Step 14: Now push the folder into the github repository using the command,

> git push -u origin main

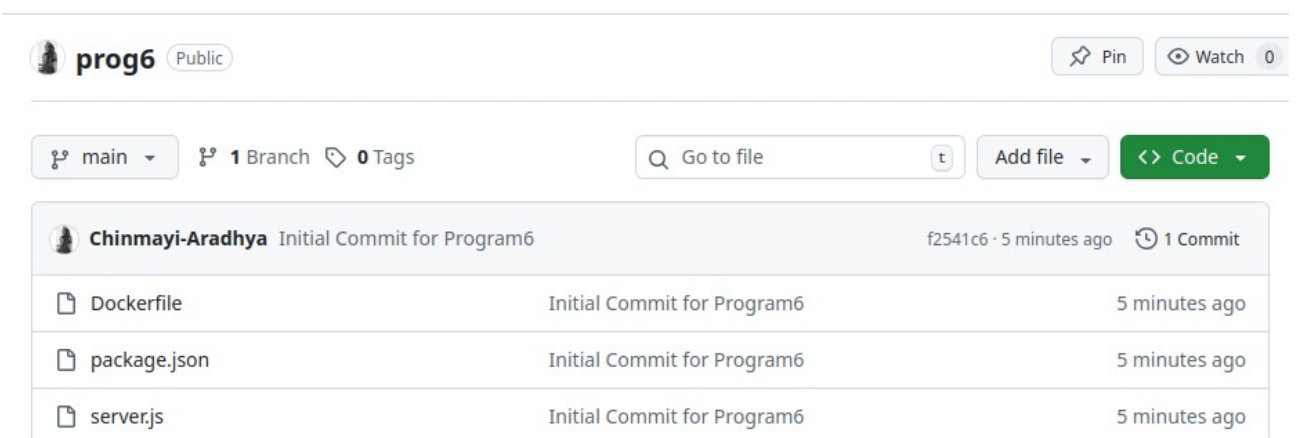
//here it will ask for the github username and password, for the password paste the PAT which we have created in step 8

```

1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog6$ git push -u origin main
Username for 'https://github.com': Chinmayi-Aradhya
Password for 'https://Chinmayi-Aradhya@github.com':
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 16 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (5/5), 989 bytes | 989.00 KiB/s, done.
Total 5 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/Chinmayi-Aradhya/prog6.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.

```

Step 15: Open the github browser page again and refresh the repository page, all the files of the folder prog6 will be displayed here



The screenshot shows the GitHub web interface for a repository named 'prog6' by user 'Chinmayi-Aradhya'. The repository is marked as 'Public'. At the top right, there are buttons for 'Pin' and 'Watch' (0). Below the repository name, it shows 'main' as the selected branch, '1 Branch', and '0 Tags'. There is a search bar 'Go to file' and buttons for 'Add file' and 'Code'. The commit history table shows one commit: 'Initial Commit for Program6' by 'Chinmayi-Aradhya' at 'f2541c6' from '5 minutes ago'. The commit message is 'Initial Commit for Program6'. The files included in the commit are 'Dockerfile', 'package.json', and 'server.js', all created '5 minutes ago'.

File	Commit Message	Time
Dockerfile	Initial Commit for Program6	5 minutes ago
package.json	Initial Commit for Program6	5 minutes ago
server.js	Initial Commit for Program6	5 minutes ago